

Complex Analysis Qualifying Exam Review

Fall 2026

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Chapter 1

Fall 2025

Problem 1.1 (Theorem 5.2, Stein, Chapter 2). Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \rightarrow f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \rightarrow \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Proof. This is an immediate consequence of Morera's theorem. Take an arbitrary triangle T contained in $\mathbb{D}$, we know for any n ,

$$\int_T f_n(z) dz = 0$$

by Goursat's theorem. By assumption, $f_n \rightarrow f$ uniformly, thus

$$\int_T f(z) dz = \lim_{n \rightarrow \infty} \int_T f_n(z) dz = 0$$

By Morera's theorem, f is holomorphic in $\mathbb{D}$.

Morera's theorem. □

Problem 1.2. 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

Proof. Use residue theorem for $z_1 = e^{\frac{\pi i}{4}}, z_2 = e^{\frac{3\pi i}{4}}$. □

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

Proof. Maximum modulus principle. □

Problem 1.4.8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Proof. Rouché's theorem.

□

Chapter 2

Cauchy's integral formula

Proposition 2.1 (Cauchy's theorem). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U , if γ is a closed smooth curve whose interior is contained in U , then

$$\int_{\gamma} f(z) dz = 0$$

Proposition 2.2 (Cauchy's integral formula). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then for any $z \in D_r(z_0)$,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw$$

where $\gamma = z_0 + re^{i\theta}$, $\theta \in [0, 2\pi]$. Moreover,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{n+1}} dw$$

Proposition 2.3 (Cauchy's inequality). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then

$$|f^{(n)}(z_0)| \leq \frac{n! \sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.4 (Liouville's theorem). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a bounded entire function, then f must be constant.

Proposition 2.5 (Holomorphic and analytic). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then f has a power series expansion at z_0 :

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D_r(z_0)$, where

$$a_n = \frac{f^{(n)}(z_0)}{n!}$$

This combined with the previous theorem gives a bound on the coefficients:

$$|a_n| \leq \frac{\sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.6 (Morera's theorem). Suppose f is a continuous function in an open set U , if for any triangle T contained in D ,

$$\int_T f(z) dz = 0$$

then f is holomorphic.

Problem 2.1. Evaluate the following integral:

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Proof. The answer is

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$$

We use the contour of a semicircle skipping $x = 0$. Let $f(z) = \frac{1 - e^{iz}}{z^2}$, then by Cauchy's integral formula, we have

$$\int_{\gamma_R} f(z) dz + \int_{\gamma_\varepsilon} f(z) dz + \int_{|z| > \varepsilon} f(z) dz = 0 \quad (1)$$

where

$$\lim_{\varepsilon \rightarrow 0} \int_{|z| > \varepsilon} f(z) dz = 2 \int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Hence it suffices to compute the first two terms in (1). Let $z = Re^{i\theta}$, we see that

$$\begin{aligned} \left| \int_{\gamma_R} f(z) dz \right| &= \left| \int_0^\pi \frac{1 - e^{iRe^{i\theta}}}{R^2 e^{2i\theta}} iRe^{i\theta} d\theta \right| \\ &\leq \int_0^\pi \frac{2}{R} d\theta \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Similarly, we can compute $\int_{\gamma_\varepsilon} f(z) dz$ by setting $z = \varepsilon e^{i\theta}$ and letting $\varepsilon \rightarrow 0$.

$$\begin{aligned} \int_{\gamma_\varepsilon} f(z) dz &= \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon^2 e^{2i\theta}} i\varepsilon e^{i\theta} d\theta \\ &= i \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}} d\theta \rightarrow -\pi \end{aligned}$$

as $\varepsilon \rightarrow 0$, since

$$\lim_{a \rightarrow 0} \frac{1 - e^{ia}}{a} = -i$$

□

Problem 2.2. Let U be a region. Suppose f is holomorphic in U , and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n , show that $f \equiv 0$.

Proof. Since $z_n \rightarrow z$, we know $f(z) = \lim_{z_n \rightarrow z} f(z_n) = 0$. Assume f is not identically zero, then writing it as a power series, we have

$$f(w) = (w - z)^n g(w)$$

where $g(w) = a_n + a_{n+1}(w - z) + a_{n+2}(w - z)^2 + \dots$, for some $a_n \neq 0$. We know $g(z) = a_n \neq 0$, hence there exists a neighborhood V around z such that $g(w) \neq 0$ whenever $w \in V$. Hence, if $w \in V$, then $f(w) = 0$ iff $w = z$. This is a contradiction to $z_n \rightarrow z$ and $f(z_n) = 0$ for all n . □



Warning 2.1. This shows that a nonzero holomorphic function has isolated zeros.

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0 , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$.

Lemma 2.1. Let f be a nonzero holomorphic function, then f cannot have uncountably many zeros.

Proof. Let $Z = \{z \in \mathbb{C} : f(z) = 0\}$, we know that Z contains no limit point. We can tile the complex plane using unit squares and suppose Z is uncountable, the intersection of Z with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. □

□

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k .

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$ because $m > k$. Hence we are done. $\square$

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Proof. (a) We write f as the Taylor series about 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

Using Cauchy's inequality, for any R large, if $n \geq 1$,

$$|f^{(n)}(0)| \leq \frac{n! \sup_{|z|=R} |f(z)|}{R^n} \rightarrow 0$$

as $R \rightarrow \infty$. Thus $f(z) = a_0$, for some $a_0 \in \mathbb{C}$.

(b) The function is not entire. $\square$

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0 , then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. $\square$

Problem 2.7 (Spring 2015, Q6). Suppose f is a nonconstant entire function, then show that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Proof. Suppose there exists $w \in \mathbb{C}$ such that for a small $r > 0$, $D_r(w) \cap f(\mathbb{C}) = \emptyset$. Then we consider the function

$$\frac{1}{f(z) - w}$$

This function is entire and bounded, implying f is constant, which is a contradiction. $\square$

Chapter 3

Singularities

Definition 3.1 (residue). If f has a pole of order n at z_0 , then

$$\operatorname{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z)$$

Proposition 3.1 (Removable singularity). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a removable singularity at z_0 iff

$$\lim_{z \rightarrow z_0} (z - z_0) f(z) = 0$$

iff there exists a neighborhood N of z_0 where f is bounded on N .

Proposition 3.2 (Pole). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a pole at z_0 iff

$$\lim_{z \rightarrow z_0} |f(z)| = \infty$$

iff there exists a non-vanishing holomorphic function h and a unique n such that

$$f(z) = (z - z_0)^{-n} h(z)$$

Proposition 3.3 (Residue theorem). Suppose f is holomorphic in an open set containing a smooth closed curve γ and its interior, except for poles at points $z_1, \dots, z_n$ inside of γ . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^n \operatorname{res}_{z_i} f$$

Proposition 3.4 (Casorati-Weierstrass). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$ and suppose f has an essential singularity at z_0 , then the image of f is dense in $\mathbb{C}$.

Proposition 3.5 (Argument principle). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If f has no poles and never vanishes on γ , then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} = (\#\text{zeros of } f \text{ inside } \gamma) - (\#\text{poles of } f \text{ inside } \gamma)$$

where zeros and poles are counted with multiplicities.

Proposition 3.6 (Rouche's theorem). Let f be a meromorphic function in an open set containing a smooth closed curve γ and its interior. If

$$|g(z)| < |f(z)| \text{ for all } z \in \gamma$$

then $f, f + g$ have the same number of zeros inside of γ .

Proposition 3.7 (Open mapping theorem). If f is holomorphic and non-constant in a region Ω , then f is open.

Proposition 3.8 (Maximum modulus principle). If f is holomorphic on a region Ω with compact closure $\bar{\Omega}$ and continuous on $\bar{\Omega}$, then

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \partial\Omega} |f(z)|$$

(There are many equivalent statements of MMP, the idea is the same: a nonconstant holomorphic function f cannot attain the maximum in the interior of the compact set).

there are more here

Proposition 3.9.

Problem 3.1. Compute

$$\int_0^{\infty} \frac{1}{1+x^2} dx$$

Proof. Let $f(z) = \frac{1}{1+z^2}$. Using a semicircle along the real axis and applying the residue theorem,

$$\int_{|x|<R} \frac{1}{1+x^2} dx + \int_{\gamma_R} f(z) dz = 2\pi i \text{res}_i f$$

where γ_R is the semicircle arc. We first show that $\int_{\gamma_R}$ goes to zero as $R \rightarrow \infty$.

$$\left| \int_{\gamma_R} \frac{1}{1+z^2} dz \right| = \left| \int_0^{2\pi} \frac{1}{1+R^2 e^{2i\theta}} i R e^{i\theta} d\theta \right| = O\left(\frac{1}{R}\right)$$

Then we compute

$$\text{res}_i f = \lim_{z \rightarrow i} (z - i) f(z) = \pi$$

Hence letting $R \rightarrow \infty$, we see

$$\int_0^{\infty} \frac{1}{1+x^2} dx = \frac{\pi}{2}$$

□

Problem 3.2. Compute the following integral: fix $0 < a < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx$$

Proof. Let $f(z) = \frac{e^{az}}{1+e^z}$. We notice that if we use the same semicircle contour then we'd enclose more and more residues, hence now we will use a rectangle contour along the following four lines: real axis, $R + it, t \in [0, 2\pi]$, $-R + it, t \in [2\pi, 0]$, and $s + 2\pi i, s \in [R, -R]$. We notice that

$$\frac{e^{a(x+2\pi i)}}{1+e^{x+2\pi i}} = e^{2a\pi i} \frac{e^{ax}}{1+e^x} = e^{2a\pi i} f(z)$$

thus denoting the two vertical lines as γ_{R+}, γ_{R-} , the residue formula implies

$$(1 - e^{2a\pi i}) \int_{|z|<R} f(z) dz + \int_{\gamma_{R+}} f(z) dz + \int_{\gamma_{R-}} f(z) dz = 2\pi i \operatorname{res}_{\pi i} f$$

We first evaluate

$$\operatorname{res}_{\pi i} f = \lim_{z \rightarrow \pi i} (z - \pi i) f(z) = -e^{a\pi i}$$

Now we show the two vertical pieces tend to 0:

$$\begin{aligned} \left| \int_{\gamma_{R+}} f(z) dz \right| &= \int_0^{2\pi} \left| \frac{e^{a(R+it)}}{1+e^{R+it}} \right| dt \\ &\leq C e^{(a-1)R} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. □

Problem 3.3. Assume that $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if Z is unbounded, then f has an essential singularity at ∞ .

Problem 3.4. Suppose f is an injective entire function, show f is linear.

Problem 3.5 (Spring 2017, Q6). Let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function where $U \subset \mathbb{C}$ is an open set containing the closure $\bar{D}$ of the unit disk

$$D = \{z \in \mathbb{C} : |z| < 1\}.$$

If $|f(z)| = 1$ for all $z \in \partial D$, then prove that

$$D \subset f(\bar{D}).$$

Proof. Argument+maximum modulus. □

Problem 3.6 (Fall 2015, Q4). Let $U \subset \mathbb{C}$ be an open set containing $\overline{D(P, r)}$. Prove that if $f : U \rightarrow \mathbb{C}$ is a holomorphic function such that f is nowhere zero on $\partial D(P, r)$ and $g : U \rightarrow \mathbb{C}$ is a holomorphic function sufficiently uniformly close to f on $\partial D(P, r)$, then the number of zeros of f in $D(P, r)$ equals the number of zeros of g in $D(P, r)$ (counting multiplicity).

Proof. Argument. □

Problem 3.7 (Spring 2021, Q1). Suppose $U \subset \mathbb{C}$ is an open set. Suppose f and g are functions holomorphic on $U \subset \mathbb{C}$. Show that if

$$|f(z)|^4 + |g(z)|^4$$

attains a maximum for some $z_0 \in U$, then f and g are constant on U .

Proof. MMP. □

Chapter 4

Conformal mapping

Definition 4.1 (automorphisms of the disk). Let $a \in \mathbb{D}$, the following function is an automorphism of the disk:

$$\phi_a(z) = \frac{z - a}{1 - \bar{a}z}$$

We have $\phi_a \circ \phi_{-a} = \text{id}$. Moreover, if f is a holomorphic function on $\mathbb{D}$, then $g(0) = 0$ if we let

$$g = \phi_{f(a)} \circ f \circ \phi_{-a}$$

Proposition 4.1 (Schwarz lemma). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$, then

1. $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
2. If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then $f(z) = cz$ for some $|c| = 1$.
3. $|f'(0)| \leq 1$ and if equality holds, then $f(z) = cz$ for some $|c| = 1$.

Proposition 4.2 (Corollary of Schwarz's lemma). If f is an automorphism of $\mathbb{D}$, then there exist $\theta \in \mathbb{R}$ and $a \in \mathbb{D}$ such that

$$f(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}$$

Namely, the only automorphisms of $\mathbb{D}$ that fix the origin are rotations.

Proposition 4.3 (Automorphisms of $\mathbb{H}$). f is an automorphism of $\mathbb{H}$ iff it is of the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}$, $ad - bc = 1$.

Problem 4.1 (Spring 2019, Q4). Show that the punctured unit disk

$$\{z : 0 < |z| < 1\}$$

and the annulus

$$\{z : 1 < |z| < 2\}$$

cannot be conformally equivalent.

Problem 4.2 (Fall 2019, Q4). Let f be a holomorphic function in the punctured disk

$$\{z : 0 < |z| < 2\}$$

satisfying

$$|f(z)| \leq \left(\log \frac{1}{|z|} \right)^{100} \quad \text{in } \{z : |z| \leq 1/2\},$$

and

$$|f(z)| = 1 \quad \text{on } |z| = 1.$$

- (a) Show that f has a removable singularity at the origin.
- (b) Show that if $f(z) \neq 0$ in $|z| < 1$, then f is constant.

Problem 4.3 (Fall 2019, Q4). Let $f(z)$ be a holomorphic function on

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\},$$

with $|f(z)| < 1$, and suppose $f(\alpha) = 0$ for some $|\alpha| < 1$. Show that for $z \in D$,

$$|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right|.$$

Proof. Schwarz lemma. □

Problem 4.4 (Spring 2016, Q5). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic function where

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

is the unit disk. Prove that if f has at least 2 fixed points then f is the identity map.

Proof. Let $z_1, z_2 \in \mathbb{D}$ be such that $f(z_1) = z_1, f(z_2) = z_2$. Then we consider the composite

$$g = \phi_{z_1} \circ f \circ \phi_{-z_1}$$

It is clear that g fixes 0, and if we let $z'_2 \neq 0$ be such that $\phi_{-z_1}(z'_2) = z_2$, then it is clear that g fixes z'_2 as well. We see that

$$g(0) = 0, \quad |g(z)| \leq 1, \quad g(z'_2) = z'_2$$

hence applying the Schwarz lemma, there exists $|c| = 1$ such that $g(z) = cz$, which implies g is constant. This shows f is constant as well. □

Chapter 5

Spring 2026

Problem 5.1. Let f be a holomorphic function on $\{z : |z| < 1\}$ that extends to a continuous function on $\{z : |z| \leq 1\}$ and maps the closed unit disk to itself. Furthermore, suppose

$$f(0) = f'(0) = \dots = f^{(n-1)}(0) = 0.$$

Show that:

(1)

$$|f(z)| \leq |z|^n \quad \text{for all } |z| \leq 1.$$

(2) If there is some $P \in D(0, 1) \setminus \{0\}$ such that

$$|f(P)| = |P|^n$$

or if

$$|f^{(n)}(0)| = n!,$$

then

$$f(z) = e^{i\theta} z^n.$$

Problem 5.2 (Also Fall 2023, Q2). Let $-1 < a < 0$. Compute

$$\int_0^\infty \frac{x^a}{1+x} dx.$$

Problem 5.3. Two parts.

(1) Let

$$f : U = \{z : 0 < |z| < 1\} \longrightarrow \mathbb{C}$$

be holomorphic and nonconstant. If there exists a sequence $\{z_n\} \subset U$ that converges to 0 and satisfies

$$f(z_n) = 0,$$

then prove that 0 is an essential singularity of f .

(2) Show that if f is an entire function that satisfies

$$\lim_{|z| \rightarrow \infty} f(z) = \infty,$$

then f is a polynomial.

Problem 5.4. Describe all injective holomorphic maps that map

$$\mathbb{C} \cup \{\infty\} \quad \text{to} \quad \mathbb{C} \cup \{\infty\}$$

and also map

$$\mathbb{R} \cup \{\infty\} \quad \text{to} \quad \mathbb{R} \cup \{\infty\}.$$

Proof. Let f be such a function, then by the Open Mapping Theorem, f is open. Now viewing $\mathbb{C} \cup \{\infty\}$ as the (compact) Riemann sphere, $f(\mathbb{C})$ is both open and closed, thus must be an automorphism. Since f also fixes $\mathbb{R} \cup \{\infty\}$, let $\mathbb{H}$ denote the upper half plane, then either $\mathbb{H} \mapsto \mathbb{H}$ or $\mathbb{H} \mapsto -\mathbb{H}$. In the first case, we know f must take the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}, ad - bc = 1$. In the second case, f can be written as the composite of the function $z \mapsto -z$ and a mobius transformation. Combined, all the maps are of the form

$$\left\{ \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{R}, ad - bc = \pm 1 \right\}$$

□